Question:
Melanie is a door-to-door saleswoman. She sold a third of her vacuum cleaners at the green house, 2 more to the red house, and half of what was left at the orange house. If Melanie has 5 vacuum cleaners left, how many did she start with?

Reasoning + Answer:

<think> To solve this problem, let's define a variable to help us work through it systematically. Let's say Melanie started with \(x \) vacuum cleaners. According to the problem, we need to follow the sequence of events she went through with her vacuum cleaners.

According to the problem, Melanie sold a third of her vacuum cleaners at the green house. So, she sold \(\frac{x}{3} \) vacuum cleaners there. This means after selling at the green house, she had \(x - \frac{x}{3} \) vacuum cleaners left. Simplifying \(x - \frac{x}{3} \) gives us \(\frac{3x}{3} - \frac{x}{3} = \frac{2x}{3} \) vacuum cleaners remaining.

Next, the problem states that she sold 2 more vacuum cleaners at the red house. So, we subtract 2 from the remaining vacuum cleaners after the green house: \(\frac{2x}{3} - 2 \).

Then, she sold half of what was left at the orange house. So, we need to find half of \(\frac{2x}{3} - 2 \). Let's denote the remaining vacuum cleaners after the orange house as \(y \). Therefore, \(y = \frac{1}{2} \left(\frac{2x}{3} - 2 \right) \).

According to the problem, after these sales, she had 5 vacuum cleaners left. So, we can set up the equation \(y = 5 \). Substituting \(y \) with the expression we found, we get: \(5 = \frac{1}{2} \left(\frac{2x}{3} - 2 \right) \).

Now, let's solve this equation for \(x \). First, we'll eliminate the fraction by multiplying both sides of the equation by 2: \(10 = \frac{2x}{3} - 2 \). Next, we'll add 2 to both sides to isolate the term with \(x \): \(10 + 2 = \frac{2x}{3} \), which simplifies to \(12 = \frac{2x}{3} \).

To solve for \(x \), we'll multiply both sides by 3 to get rid of the denominator: \(12 \times 3 = 2x \), which simplifies to \(36 = 2x \). Finally, we'll divide both sides by 2 to solve for \(x \): \(\frac{36}{2} = x \), so \(x = 18 \). </think>
<answer> 18 </answer>